

Code Decode Lab - Hands-On Python Tutorial

Lab Objective

Build a utility to encode/decode multilingual strings into bytes and back using Python.

Key Concepts

- Strings are human-readable (Unicode).
- Bytes are used by computers to store/transmit text.
- Encoding turns a string into bytes.
- Decoding turns bytes back into a string.

Part 1: Word List

```
words = ['Cafe', 'Privet', 'Konnichiwa', 'Marhaban', 'Shalom', 'Namaste', 'Annyeonghaseyo']
```

Part 2: Encode & Decode Functions

```
def encode_word(word, encoding='utf-8'):
    return word.encode(encoding)

def decode_word(byte_data, encoding='utf-8', errors='strict'):
    return byte_data.decode(encoding, errors=errors)
```

Test Example (Use non-ASCII characters)

```
test_word = '?????' # Japanese for Hello
encoded = encode_word(test_word)
print('Original:', test_word)
print('Encoded (bytes):', encoded)
decoded = decode_word(encoded)
print('Decoded (string):', decoded)
```

Expected Output

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Original: ?????

Encoded (bytes): b'\xe3\x81\x93\xe3\x82\x93\xe3\x81\xab\xe3\x81\xa1\xe3\x81\xaf'

Decoded (string): ?????